Towards the Design of an FGF Signaling Agonist

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BACKGROUND

WHAT IS FGF SIGNALING?

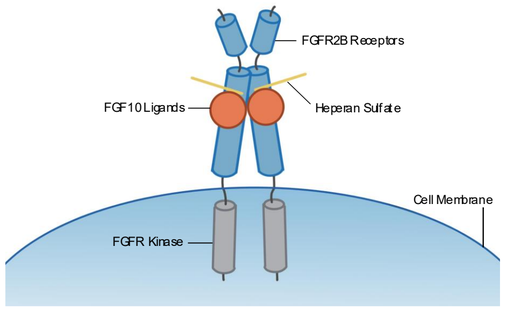
FGF signaling is a major developmental pathway involved in organ development. The Brownfield Lab is focusing on its role in developing the lung's alveolar epithelium

FGFR2 LIGAND-RECEPTOR INTERACTION

Ligands, such as Fgf10, bind to their receptor, Fgfr2, to mediate signaling.

- Signaling complex with a 2-2-2 stoichiometry
- Co-localization of internal kinases leads to downstream signaling

Figure 1: FGF Signaling Components

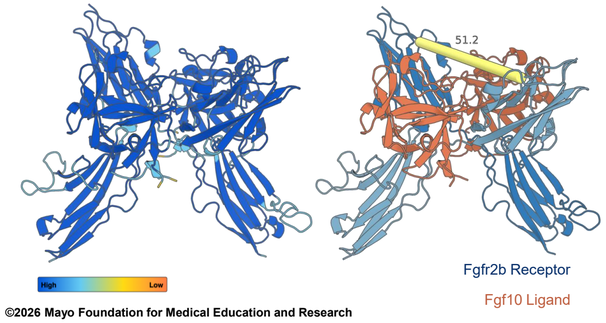


DESIGNING A FGF SIGNALING AGONIST

Trapping Fgf10 ligands into a pro-signaling conformation will likely boost signaling, so we propose turning Fgf10 into a rigidly constrained homodimer primed for receptor binding

1. Design obligate homodimers that fit the size and orientation constraints of the space between the Fgf10 ligands
2. Extend the native Fgf10 ligands into high performing homodimers to activate the Fgfr2 receptors by rigidly trapping them in a pro-signaling conformation

Figure 2: Fgfr2b Receptor + Fgf10 ligands



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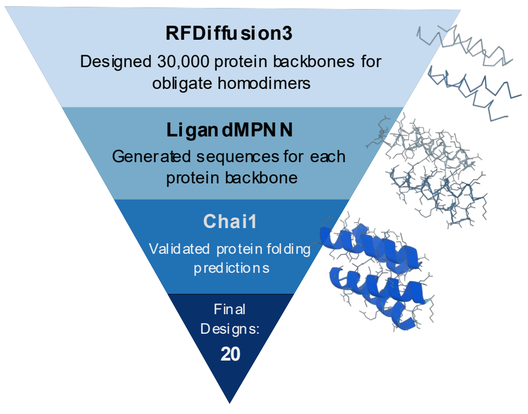
METHODS

DESIGN HOMODIMERS *IN SILICO*

Create *de novo* homodimers that may be able to lock the Fgfr2 in a pro-signaling conformation:

1. Generate protein backbones via RFDiffusion3
2. Generate homodimer sequences via LigandMPNN
3. Validate structures with Chai1
4. *In silico* evaluation to select the 20 homodimer designs for *in vitro* testing

Figure 3: Protein Design Workflow

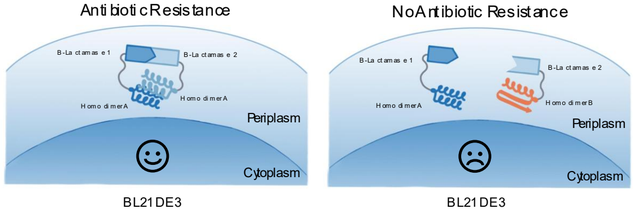


TEST HOMODIMERS *IN VITRO*

Perform a live-dead screen to evaluate homodimers:

1. Use a split beta-lactamase protein complementation assay to evaluate *de novo* homodimer binding specificity and strength
2. Strongly associating homodimers should reconstitute the beta-lactamase fragments, thus conferring antibiotic resistance and survival
3. Evaluate homodimer specificity via combinatorial testing → 400 homodimer structure complexes

Figure 4: *In Vitro* Screen



RESULTS

Figure 5: Filtered vs. Final Design Metrics

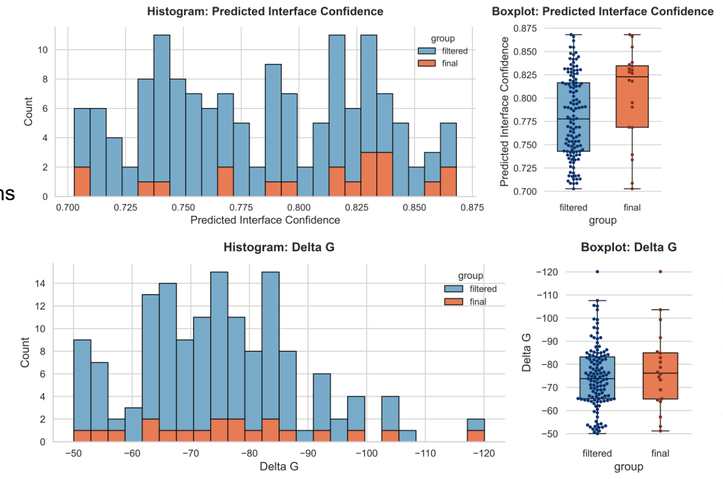


Figure 6: Sample Final Homodimers

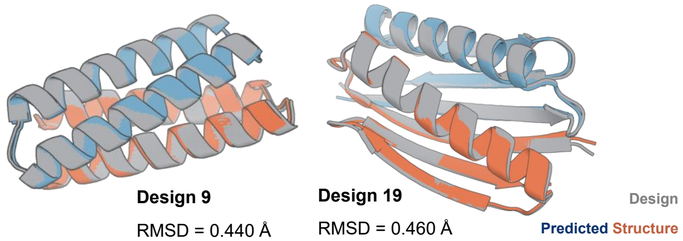
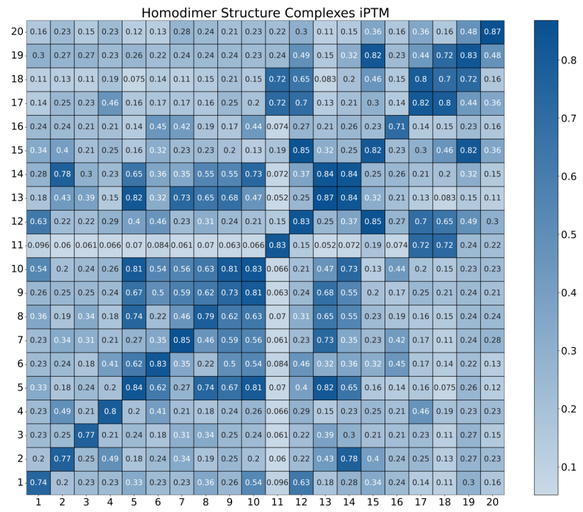


Figure 7: iPTM Heatmap



DISCUSSION

Successfully designed homodimers *in silico*

- Started with 30,000 protein backbones → Down-selected based on size and directionality
- Designed 4 sequences per backbone
- Validated by predicting structures for those sequences
- Filtered to 125 designs based on interface characteristics
- Selected 20 final designs

Comprehensively evaluated the interactions between all homodimer combinations *in silico* via iPTM metrics

- Cognate homodimer generally have the most confident interactions
- Some designs have off target interactions → Models create idealized structures, so designs share a lot of structural similarities

FUTURE DIRECTIONS

- Complete the screen to evaluate homodimers *in vitro* (in-progress)
- Extend the Fgf10 ligand into a high performing obligate homodimer to fix the FGF complex in a pro-signaling conformation
- Potential clinical applications for improving alveolar health after injury

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